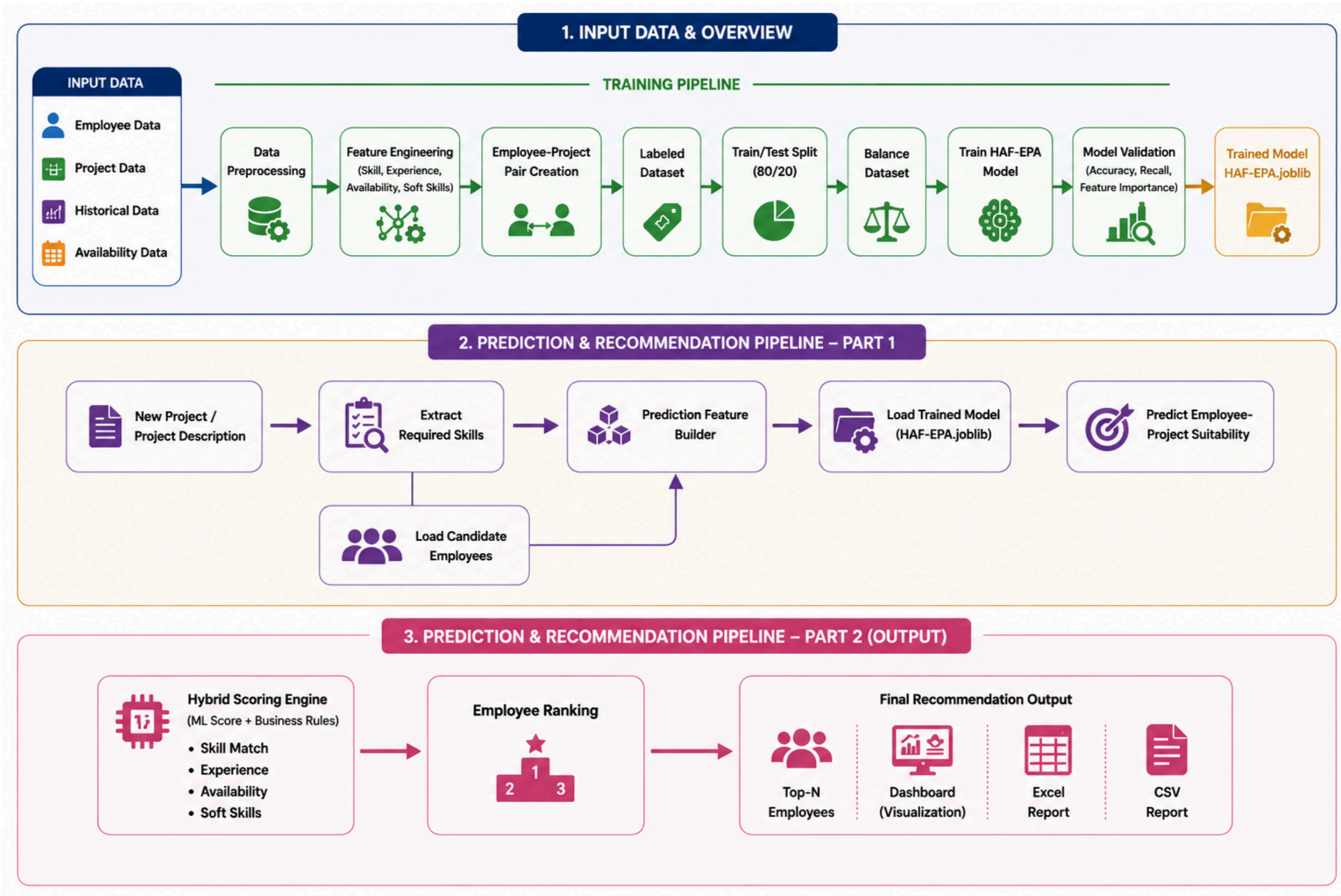

Figure X illustrates the complete workflow of the HAF-EPA (Hybrid AI Framework for Employee Project Allocation) system, which consists of three major phases: data preparation and model training, prediction generation, and recommendation generation.

Initially, employee, project, historical, and availability data are collected and processed through data preprocessing and feature engineering stages. The processed data is then used to create employee-project pairs and generate a labeled dataset for machine learning. After performing train-test splitting and dataset balancing, the HAF-EPA model is trained and validated to ensure reliable prediction performance. The trained model is subsequently saved for future use.

In the prediction phase, a new project description is analyzed to extract the required skills and project requirements. Candidate employees are loaded from the employee reference dataset, and prediction features are generated using the same feature engineering process applied during training. The trained HAF-EPA model then predicts the suitability of each employee for the given project.

Finally, in the recommendation phase, a hybrid scoring engine combines machine learning prediction scores with additional business rules, including skill matching, experience, availability, and soft-skill compatibility. Based on the final scores, employees are ranked and the most suitable candidates are identified. The system produces the final recommendation results in multiple formats, including a web-based dashboard, Excel reports, CSV reports, and a ranked Top-N employee recommendation list. This workflow enables intelligent, data-driven employee allocation and project staffing decisions.